

Walmart Sales Data Analysis Project Report

#1. Total Revenue Generated

- **Objective:** Calculate the cumulative gross revenue across all historical transactions.
- **SQL Query:**
- SQL

```
SELECT  
  ROUND(SUM(unit_price * quantity), 2) AS total_revenue  
FROM walmart;
```

- **Output:** * **total_revenue: 1209726.38**
- **Insight:** The store has hit an aggregate sales volume of approximately \$1.21 million, setting a baseline financial scale for further performance benchmarking.

#2. Total Sales by City

- **Objective:** Identify geographic revenue contribution across different operating locations.
- **SQL Query:**
- SQL

```
SELECT  
  City,  
  ROUND(SUM(unit_price * quantity), 2) AS revenue  
FROM walmart  
GROUP BY City  
ORDER BY revenue DESC;
```

- **Output:** * **Weslaco: 46351.79**
 - **Waxahachie: 40703.33**
 - **Plano: 25688.34***(Truncated list of 98 rows)*
- **Insight:** Weslaco and Waxahachie emerge as the primary regional drivers of top-line revenue, outperforming secondary markets like Plano by a wide margin.

#3. Best Selling Product Category

- **Objective:** Determine the customer demand profile by aggregating total inventory units sold across categories.

- **SQL Query:**
- SQL

```
SELECT
    category,
    SUM(quantity) AS total_quantity_sold
FROM walmart
GROUP BY category
ORDER BY total_quantity_sold DESC;
```

- **Output:** * Fashion accessories: 9653
 - Home and lifestyle: 9610
 - Electronic accessories: 1494
- **Insight:** Consumer volume is overwhelmingly dominated by Fashion and Home goods, moving significantly more units compared to higher-margin electronics.

#4. Average Rating by Category

- **Objective:** Evaluate product health and customer satisfaction metrics across product divisions.
- **SQL Query:**
- SQL

```
SELECT
    category,
    ROUND(AVG(rating), 2) AS avg_rating
FROM walmart
GROUP BY category
ORDER BY avg_rating DESC;
```

- **Output:** * Food and beverages: 7.11
 - Health and beauty: 7.00
 - Sports and travel: 6.92
- **Insight:** Food, beverage, and health products maintain the highest overall customer approval ratings, whereas lifestyle items fall slightly behind.

#5. Most Preferred Payment Method

- **Objective:** Analyze checkout trends to optimize point-of-sale friction and digital integrations.
- **SQL Query:**
- SQL

```
SELECT
    payment_method,
    COUNT(*) AS total_transactions
FROM walmart
GROUP BY payment_method
ORDER BY total_transactions DESC;
```

- **Output:** * Credit card: 4256
 - Ewallet: 3881

- **Cash:** 1832
- **Insight:** Digital mechanisms (Credit Card and E-wallets) account for the massive majority of user checkouts, reflecting low reliance on traditional cash transactions.

#6. Branch with Highest Revenue

- **Objective:** Pinpoint the single highest-performing retail hub within the business framework.
- **SQL Query:**
- SQL

```
SELECT
  Branch,
  ROUND(SUM(unit_price * quantity), 2) AS revenue
FROM walmart
GROUP BY Branch
ORDER BY revenue DESC
LIMIT 1;
```

- **Output:** * **WALM009:** 25688.34
- **Insight:** Branch **WALM009** stands out as the single highest revenue-generating asset, serving as a operational benchmark for other branches.

#7. Monthly Revenue Analysis

- **Objective:** Map sequential revenue flow across months to identify seasonal spikes or lulls.
- **SQL Query:**
- SQL

```
SELECT
  MONTH(date) AS month_no,
  MONTHNAME(date) AS month_name,
  ROUND(SUM(unit_price * quantity), 2) AS revenue
FROM walmart
GROUP BY month_no, month_name
ORDER BY month_no;
```

- **Output:** * **January:** 137481.16
 - **February:** 122667.88
 - **March:** 131692.34
 (Truncated list of 12 rows)
- **Insight:** The business maintains a steady, reliable baseline of over \$120k/month early in the fiscal year, with minor fluctuations across consecutive quarters.

#8. Top 5 Highest Value Transactions

Objective: Isolate outlier bulk orders or high-ticket ticket purchases to understand big-spender behaviors.

SQL Query:

SQL

```
SELECT
    invoice_id, City, category, quantity, unit_price,
    (quantity * unit_price) AS total_bill
FROM walmart
ORDER BY total_bill DESC
LIMIT 5;
```

Output: * Invoice 351 (McKinney): 993.00

- Invoice 168 (Edinburg): 989.80
- Invoice 558 (Bryan): 985.20

Insight: The highest single-basket transactions consistently cap out just under \$1,000, frequently driven by large quantities (maxed out at 10 items) of fashion and home goods.

#9. Average Profit Margin by Category

- **Objective:** Verify which segments return the most yield per item for strategic financial adjustments.
- **SQL Query:**
- SQL

```
SELECT
    category,
    ROUND(AVG(profit_margin), 2) AS avg_profit_margin
FROM walmart
GROUP BY category
ORDER BY avg_profit_margin DESC;
```

- **Output:** * Health and beauty: 0.40
 - Food and beverages: 0.40
 - Electronic accessories: 0.39
- **Insight:** Profit margins stay exceptionally uniform across the board at around 38% to 40%, meaning revenue generation is the true key factor in total profitability rather than disparate pricing markups.

#10. Most Profitable Category

- **Objective:** Calculate cumulative dollar profit (Revenue $\times$ Profit Margin) to find the absolute business engine.
- **SQL Query:**

```
SELECT
    category,
    ROUND(SUM(unit_price * quantity * profit_margin), 2) AS profit
FROM walmart
GROUP BY category
ORDER BY profit DESC;
```

- **Output:** * Fashion accessories: 192314.89
 - Home and lifestyle: 192213.64
 - Electronic accessories: 30772.49
- **Insight:** Driven by massive scale and volume, Fashion and Lifestyle items bring in the most pure dollar profit for Walmart, vastly outpacing all other divisions.

#11. Peak Shopping Hours

- **Objective:** Map customer traffic volume by the hour of the day to determine peak operational periods.
- **SQL Query:**

```
SELECT

    HOUR(time) AS hour_of_day,

    COUNT(*) AS transactions

FROM walmart

GROUP BY hour_of_day

ORDER BY transactions DESC;
```

- **Output:** * 15: 1191 16: 1173 17: 1027
- **Insight:** Customer traffic spikes drastically in the late afternoon (3:00 PM to 5:00 PM), suggesting the optimal window for floor staff scheduling and flash promotions.

#12. Revenue by Payment Method

- **Objective:** Correlate total financial intake with preferred customer checkout methods.
- **SQL Query:**

```
SELECT
```

```
payment_method,  
  
ROUND(SUM(unit_price * quantity), 2) AS revenue  
  
FROM walmart  
  
GROUP BY payment_method  
  
ORDER BY revenue DESC;
```

- **Output:** * Credit card: 488821.02 Ewallet: 457316.07 Cash: 263589.29
- **Insight:** Digital payment methods dominate total revenue generation, proving that card and digital wallet users contribute significantly larger ticket sizes than cash users.

#13. Which City Gives the Highest Average Rating?

- **Objective:** Uncover geographic variations in customer satisfaction levels.
- **SQL Query:**
- SQL

```
SELECT  
  
City,  
  
ROUND(AVG(rating), 2) AS avg_rating  
  
FROM walmart  
  
GROUP BY City  
  
ORDER BY avg_rating DESC;
```

- **Output:** * Austin: 7.00 Huntsville: 6.81 Pflugerville: 6.73
- **Insight:** Austin leads customer satisfaction with the highest average baseline rating, while other regional hubs remain highly competitive within a tight scoring window.

#14. Rank Cities by Revenue (Window Function)

- **Objective:** Use SQL window functions to assign sequential performance ranks to operating cities based on total sales volume.
- **SQL Query:**

```
SELECT  
  
City,  
  
revenue,  
  
RANK() OVER (ORDER BY revenue DESC) AS city_rank
```

```

FROM

(

    SELECT

        City,

        SUM(unit_price * quantity) AS revenue

    FROM walmart

    GROUP BY City

) t;

```

- **Output:** * **Weslaco: 46351.79 Waxahachie: 40703.33 Plano: 25688.34**
- **Insight:** High-tier cities like Weslaco and Waxahachie securely claim the top ranking spots, creating a clear gap above secondary urban markets.

#15. Find the Most Popular Category in Each City

- **Objective:** Determine hyper-local consumer demand profiles by cross-referencing top categories across distinct regions using dynamic window partitioning.
- **SQL Query:**
- SQL

```

WITH category_sales AS (

    SELECT

        City,

        category,

        SUM(quantity) AS total_qty,

        RANK() OVER (

            PARTITION BY City

            ORDER BY SUM(quantity) DESC

        ) AS rnk

    FROM walmart

    GROUP BY City, category

)

```

SELECT

City,

category,

total_qty

FROM category_sales

- **Output:** * Abilene: Fashion accessories (82) Alamo: Home and lifestyle (79) Alice: Fashion accessories (60)
- **Insight:** Consumer demand is highly localized, with Fashion Accessories and Home & Lifestyle split across territories, highlighting where to selectively shift regional supply line priorities.